



HUMBER BAY PARK EAST MAJOR MAINTENANCE PROJECT

Concept Brief

February 2021

Table of Contents

1	INTRODUCTION.....	0
2	SITE DESCRIPTION.....	1
2.1	Location	1
2.2	Land Usage	1
2.3	Existing Erosion Control Structures	2
2.3.1	Western Armourstone Headland (WF11.10)	3
2.3.2	Cobble/Gravel Beach (WF11.11).....	6
2.3.3	Eastern Armourstone Headland (WF11.12).....	8
2.4	Typical Slope/Channel Dimensions	9
3	BASELINE INVENTORY	11
4	PERMITS.....	11
5	CONCEPT DESIGNS.....	11
5.1	WF11.10 – Western Headland	12
5.2	WF11.11 – Cobble/Rubble Beach.....	16
5.3	WF11.12 – Eastern Headland	17

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PROJECT NAME	Humber Bay Park East Major Maintenance Project
REGION	Etobicoke, City of Toronto
TRCA DIVISION	Restoration and Infrastructure (Engineering Projects)
CFN	61338

1 INTRODUCTION

Humber Bay Park is a waterfront park that is located at the mouth of Mimico Creek on the north shore of Lake Ontario. The park consists of two peninsulas (Humber Bay Park East and Humber Bay Park West) that extend out into Lake Ontario. Park construction began in the 1970s as a lake-filling operation which reached its current form in the 1990s. The parklands are armoured with a series of headlands and beaches to protect the shoreline from erosion. The park is a large urban green space, featuring wetlands, forests, and meadows, as well as a number of cultural and natural heritage features including park trails, a marina, the Air India Memorial, and the Humber Bay Butterfly Habitat Project. The park is heavily used by the local community and this usage is expected to increase as the neighbourhood evolves into a denser, high-rise residential community.

Erosion control structures, including headlands and beaches, were built in the early 1980s in order to protect the shoreline. Toronto and Region Conservation Authority's (TRCA) Erosion Risk Management Program (ERMP) has monitored these structures on an annual basis since 2006. Both the western armourstone headland (WF11.10) and the cobble/gravel beach (WF11.11) are currently classified as in overall "fair" condition, but with a structural stability rating of "failing" due to several deficiencies. The eastern armourstone headland (WF11.12) is classified as in overall "poor" condition with a structural stability rating of "failing" due to its deficiencies.

In the fall of 2016, Riggs Engineering Limited undertook a comprehensive assessment of the 73 TRCA-owned shoreline erosion control structures between Marie Curtis Park in the west and the western extent of Ontario Place in the east. Based on the ranking system accounting for structure condition, adjacent land use function, environmental factors, and others, Humber Bay Park (East and West) ranked fifth overall in priority. Since the comprehensive assessment took place, the deterioration of these structures has been exacerbated by the 2017 and 2019 high lake level events as well as the April 14-15, 2018 wind storm. Major maintenance is required to return these structures to a state of good repair and provide long-term protection to the Humber Bay Park East shoreline.

2 SITE DESCRIPTION

2.1 Location

The site is located in Humber Bay Park East, along the north shore of Lake Ontario in Etobicoke, City of Toronto. The closest major intersection is Lake Shore Boulevard West and Park Lawn Road with the park accessible via Marine Parade Drive. The general location of the site and erosion control structures are shown below in **Figure 1**.

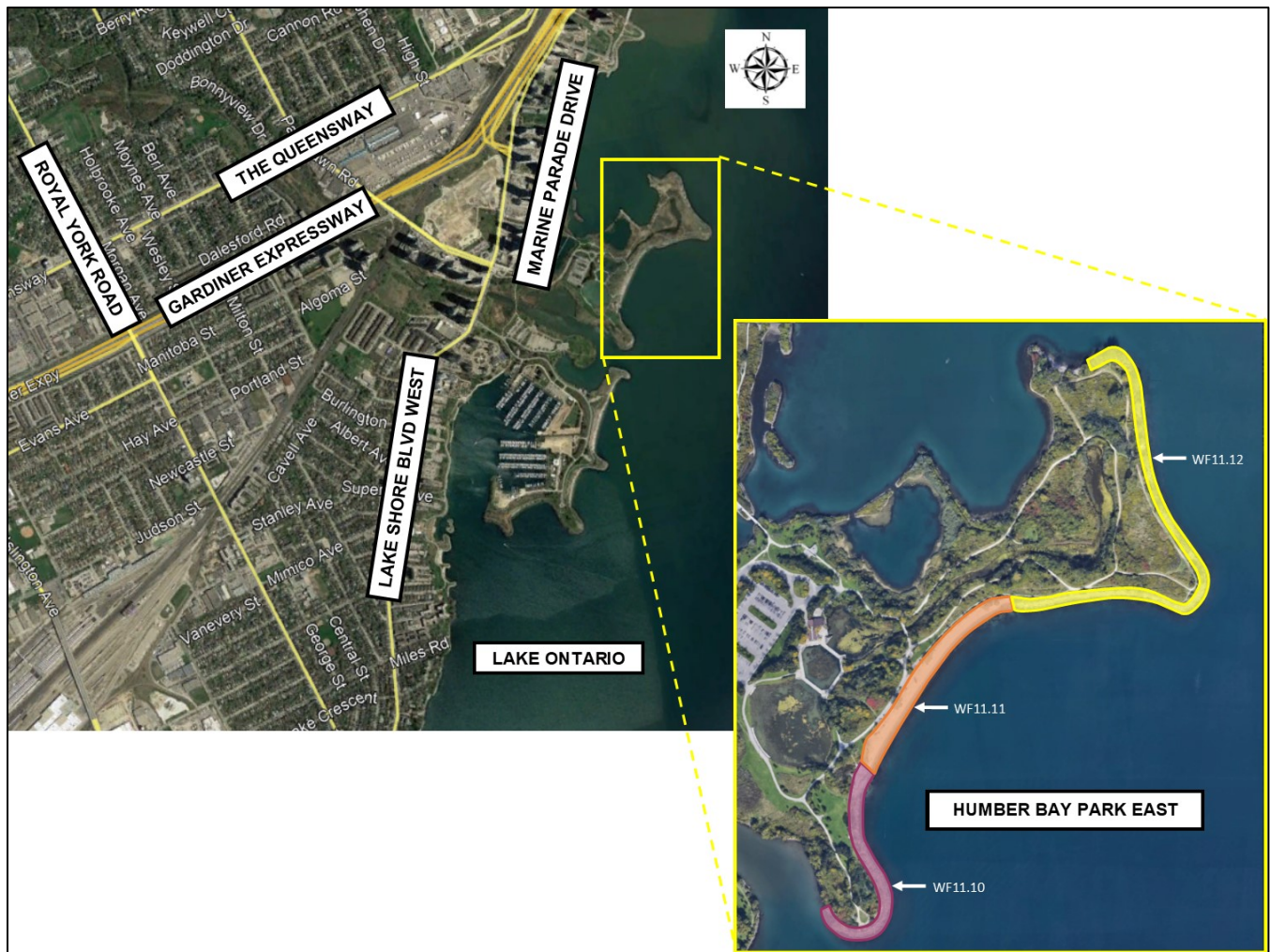


Figure 1. Map of Humber Bay Park East and location of erosion control structures. Source: TRCA, 2019.

2.2 Land Usage

The area adjacent to the structures is parkland and includes formal and informal trails that run parallel to the shoreline. The park features a system of streams and ponds, one of which drains at the eastern limit of the western headland (WF11.10) as shown in **Figure 2**.



Figure 2. Location of pond drainage at eastern limit of Armourstone Headland WF11.10. Source: TRCA, 2021.

2.3 Existing Erosion Control Structures

The condition of each of the three erosion control structures within the project area is summarized in **Table 1**. Photos and sketches of the structures are presented in **Figure 3** through **Figure 12**.

Table 1. Overall condition of the erosion control structures included in the Humber Bay Park East Major Maintenance Project. Source: TRCA, 2020.

Structure	Structural Stability	Overall Condition	Deficiencies
WF11.10 (Western Armourstone Headland)	Failing	Fair	- Minor displacement of armourstone along western and central extents of structure - Scouring of areas above the headland due to overtopping – some rip rap has been placed in key areas for temporary short-term protection (Figure 3) - Backshore scouring placing park paths and benches at risk - Irregular backshore profile – some areas are at risk of flooding during periods of high water levels (Figure 4) - Major displacement of armourstone at eastern extent of structure where pond drainage occurs (Figure 5) - Voids between armourstones

			See Figure 6 for sketch
WF11.11 (Gravel/Rubble Beach)	Failing	Fair	- Slumping, undercutting, and scouring along the backshore (Figure 7) - Scouring has encroached on an informal path - Minor debris buildup throughout the structure - Complete displacement of material and backshore scouring at western limit of the structure (Figure 8) - See Figure 9 for sketch
WF11.12 (Eastern Armourstone Headland)	Failing	Poor	- Minor armourstone and rip rap displacement at toe throughout the structure (Figure 10) - Small areas of major armourstone displacement - Multiple scours throughout the structure (some filled with rip rap but are still encroaching on park paths) (Figure 11) - Minor woody debris buildup in some areas - See Figure 12 for sketch

2.3.1 Western Armourstone Headland (WF11.10)



Figure 3. Area of backshore scouring due to overtopping. Temporary rip rap treatment installed in damaged area. Source: TRCA 2019.



Figure 4. Irregular armourstone crest profile with areas of flooding behind the structure during the high Lake Ontario water level event in 2019. Source: TRCA, 2019.



Figure 5. Condition of eastern limit of the structure with visible displacement of armourstone material. The pond drainage can be seen behind the armourstone. Source: TRCA, 2019.

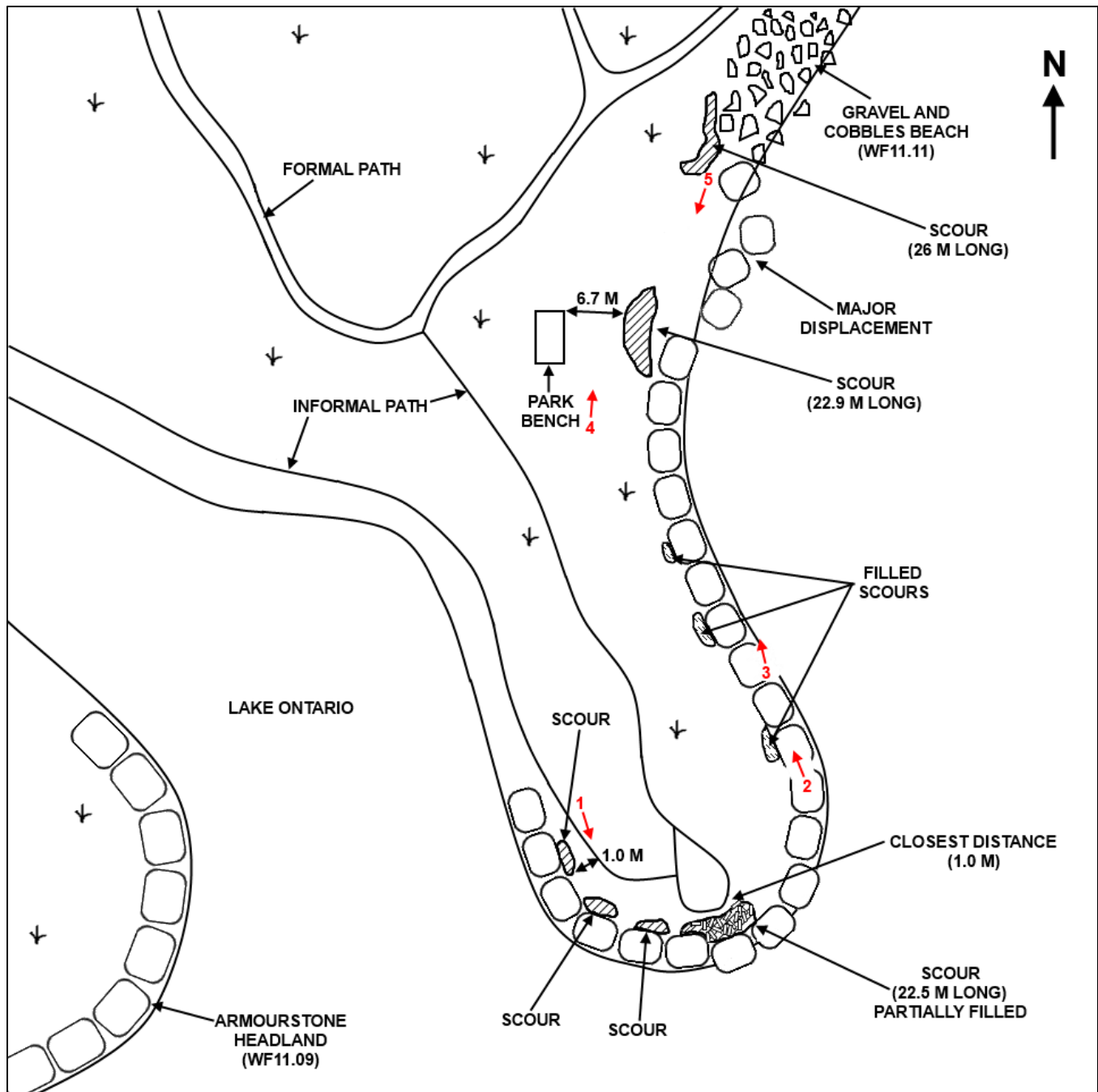


Figure 6. Sketch of WF11.10 deficiencies as identified during the 2020 ERMP inspection. Source: TRCA, 2020.

2.3.2 Cobble/Gravel Beach (WF11.11)



Figure 7. Condition of the cobble/gravel beach (WF 11.11), with visible scouring and undercutting. Source: TRCA, 2019.

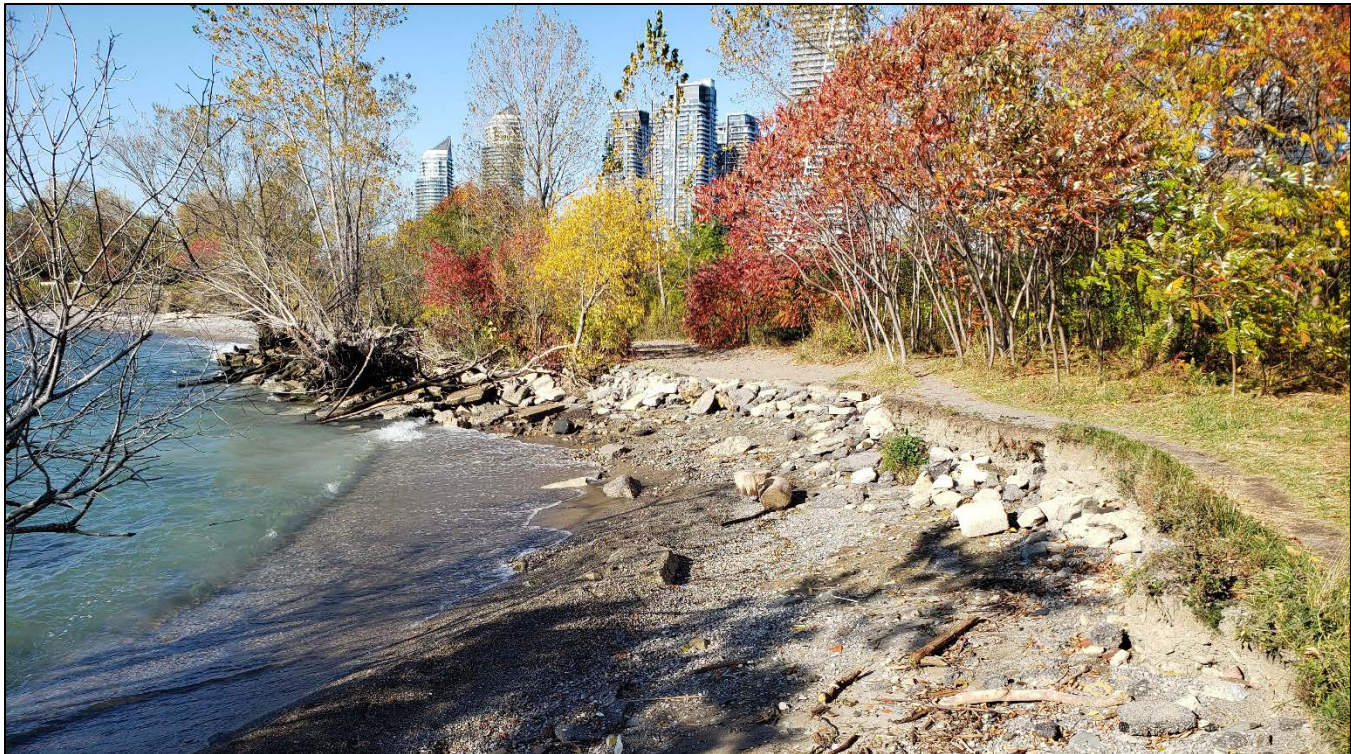


Figure 8. Scouring and material displacement encroaching on informal trail at eastern limit of structure. Source: TRCA, 2019.

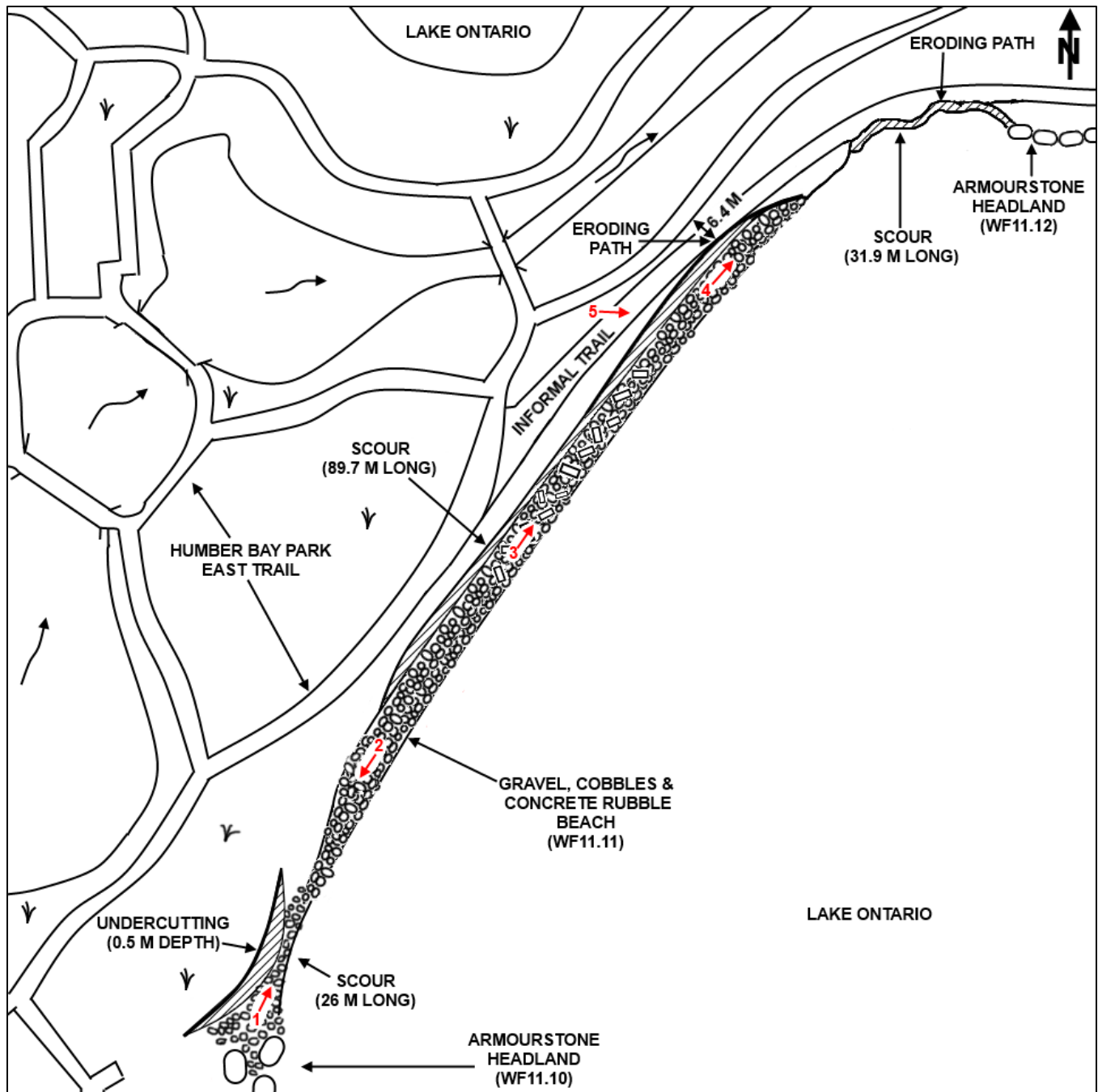


Figure 9. Sketch of WF11.11 deficiencies as identified during the 2020 ERMP inspection. Source: TRCA, 2020.

2.3.3 Eastern Armourstone Headland (WF11.12)

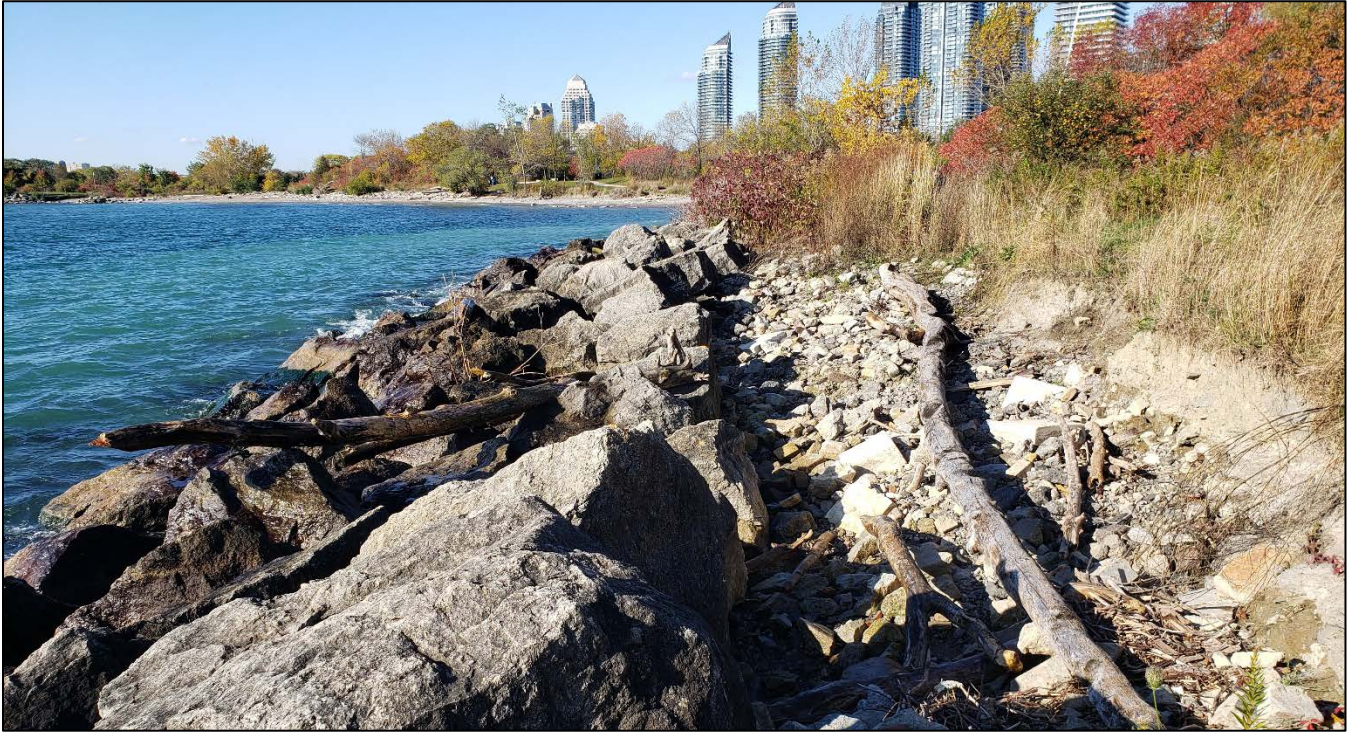


Figure 10. Condition of the eastern armourstone headland (WF11.12), with visible displacement of material, backshore erosion, and scouring. Source: TRCA, 2019.



Figure 11. Scouring and armourstone displacement along eastern section of structure WF11.12. Source: TRCA, 2019.

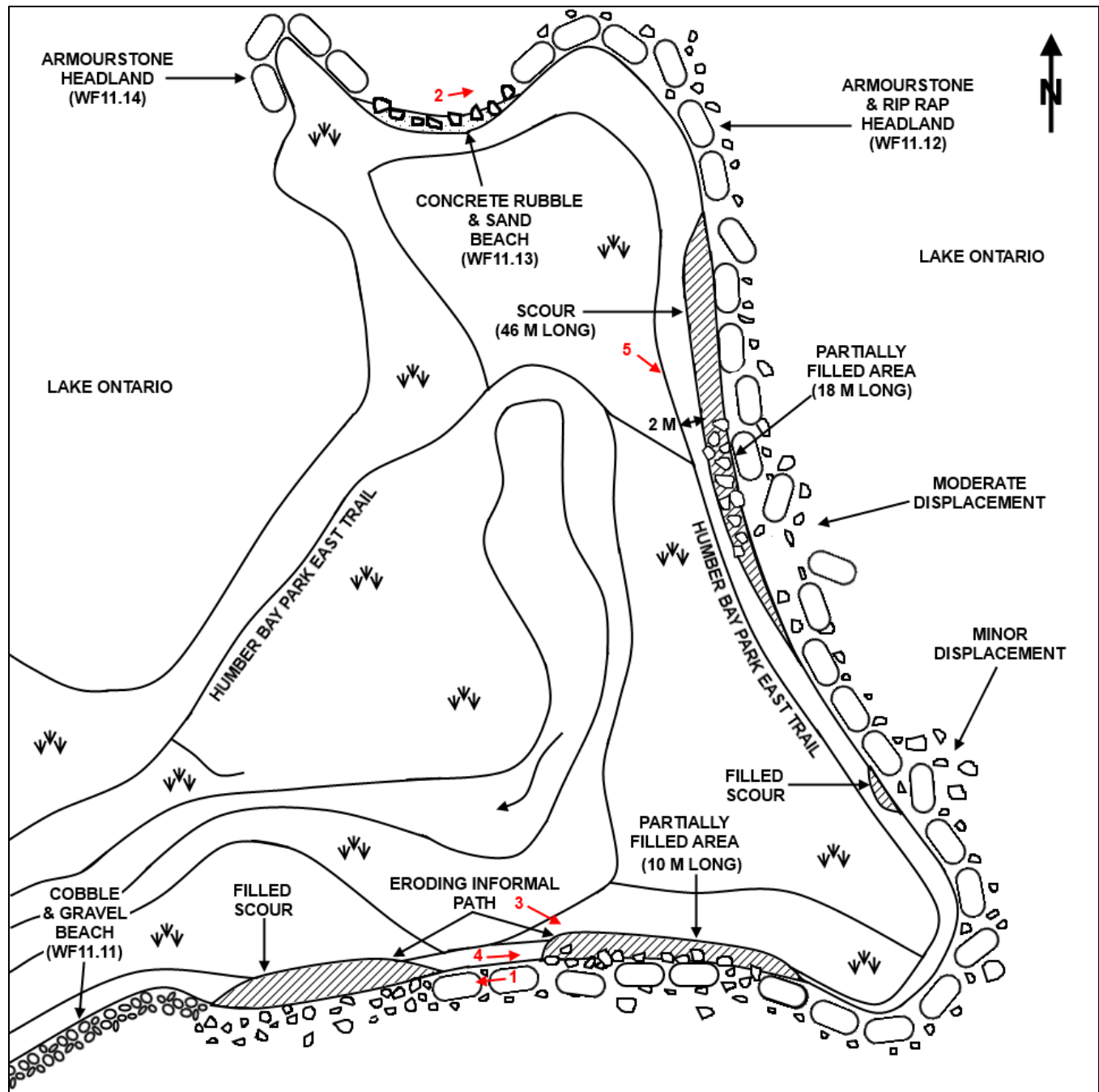


Figure 12. Sketch of WF11.12 deficiencies as identified during the 2020 ERMP inspection. Source: TRCA, 2020.

2.4 Typical Slope/Channel Dimensions

The shoreline erosion control structures extend approximately 2 m above the waterline. The crest of the structures are around 77 metres above-sea-level (m.a.s.l.) with most of the park at an elevation of about 78 m.a.s.l. The typical near shore depth is approximately 72-73 m.a.s.l. A contour map of the park is shown in **Figure 13**.



Figure 13. Contour map for Humber Bay Park East. Source: TRCA, 2019.

3 BASELINE INVENTORY

Baseline data has been collected to support conceptual design development. A summary of the baseline conditions reviewed is presented in **Table 2**. A more detailed review of existing conditions will be undertaken in support of the detailed design phase and this information will be presented in the future Detailed Design Project Brief for circulation.

Table 2. Summary of baseline inventory conducted for the Humber Bay Park East Major Maintenance Project.

Category	Status	Notes
Archaeology	Cleared	- Stage 1 Complete - Stage 2 determined to be unnecessary
Aquatic Inventory and Habitat	Complete	- Three monitoring stations used - Most common species found: Alewives, White Suckers, Rainbow Smelt
Other Projects	In Progress	- Pond Revitalization Project - Humber Bay Park Master Plan guidelines - Will coordinate with any other ongoing works as necessary
Soil Characterization	Complete	- Conducted in Spring 2020 for concurrent pond revitalization project - Soil contaminants were found in various concentrations throughout the pond study area - A Soil Management Plan may be needed if components of the Project require soil excavation (including tree plantings)
Species at Risk	Complete	- Site preparation should start at the end of July to avoid disrupting nesting songbirds - Construction activity should be maintained at a high level from April-June to discourage birds from nesting during lulls in construction and possible nest abandonment - There is a possibility for map turtles to nest in sandy/gravelly areas (though unlikely due to high amount of human activity in the park). If nests are discovered, appropriate protection from predators should be taken
Topographic and Bathymetric Survey	In Progress	- Topographic and bathymetric survey completed to inform designs - Additional survey may be required (i.e.: staging areas)
Tree Inventory	Complete	Most common tree species found were Carolina Poplar, Green Ash, Siberian Elm, Manitoba Maple, and European Alder

4 PERMITS

The permits and approvals process will commence during the detailed design phase. All applicable permits and approvals will be obtained prior to implementation.

5 CONCEPT DESIGNS

The overall approach to the concept designs aims to prioritize long-term resiliency while reducing the need for excavation and off-site disposal in order to limit disturbance and impacts of the construction activities while

addressing the leading erosion concerns of the park, including low crest elevation and wave overtopping. The proposed concepts have been designed to address these concerns by placing new material over the existing structures where toe and slope stability permits, thereby minimizing excavation requirements. Two sections have been recommended to undergo a full reconstruction where damage to the shoreline is extensive and the existing material is not stable enough to act as core material. The concepts were developed using a design water level of 75.8 m.a.s.l. and were designed to ensure that the overtopping rate is below 10 litres/second.

5.1 WF11.10 – Western Headland

The treatment outlined by the concept designs for the western headland includes the placement of fill over the existing profile. The crest elevations at the western extent will be raised to 77 m.a.s.l. using 300-600mm rip rap and will transition to a 3-5 tonne armourstone treatment with a crest elevation of 78.5 m.a.s.l. at the southern extent of the structure. The armourstone treatment will transition to an elevation of 77.8 m.a.s.l. at the eastern extent of the headland. Please see the plan view and cross-sections in **Figure 14** and **Figure 15**. Armourstone sizes vary based on the requirements of the treatment; the more exposed areas on the southern tip of the structure require 3-5 tonne armourstone, while more sheltered areas near the eastern limit require 2-3 tonne armourstone.

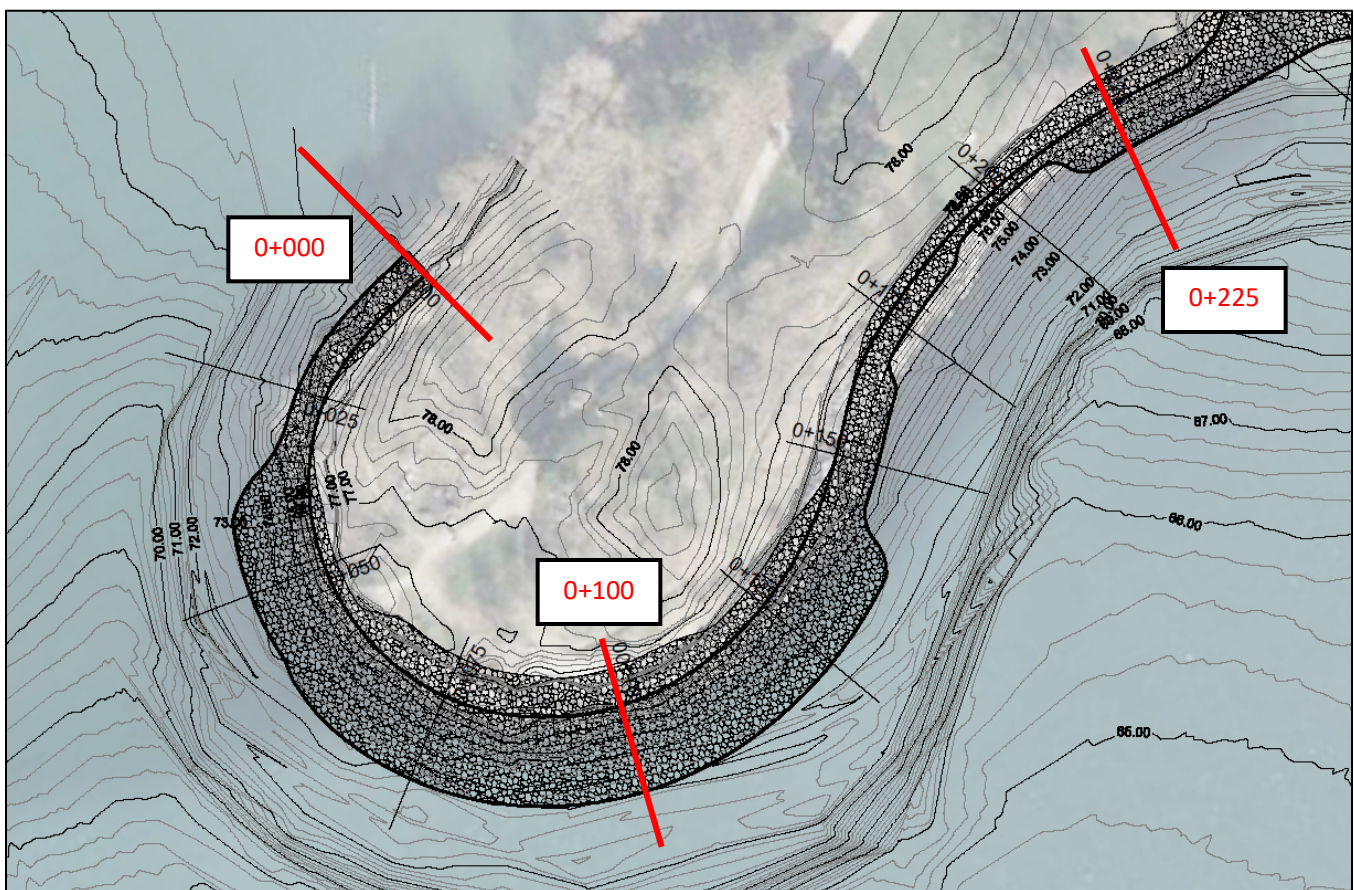


Figure 14. Planview of concept designs for structure WF11.10. Red lines denote cross-sections shown in later figures. Source: Shoreplan, 2020.

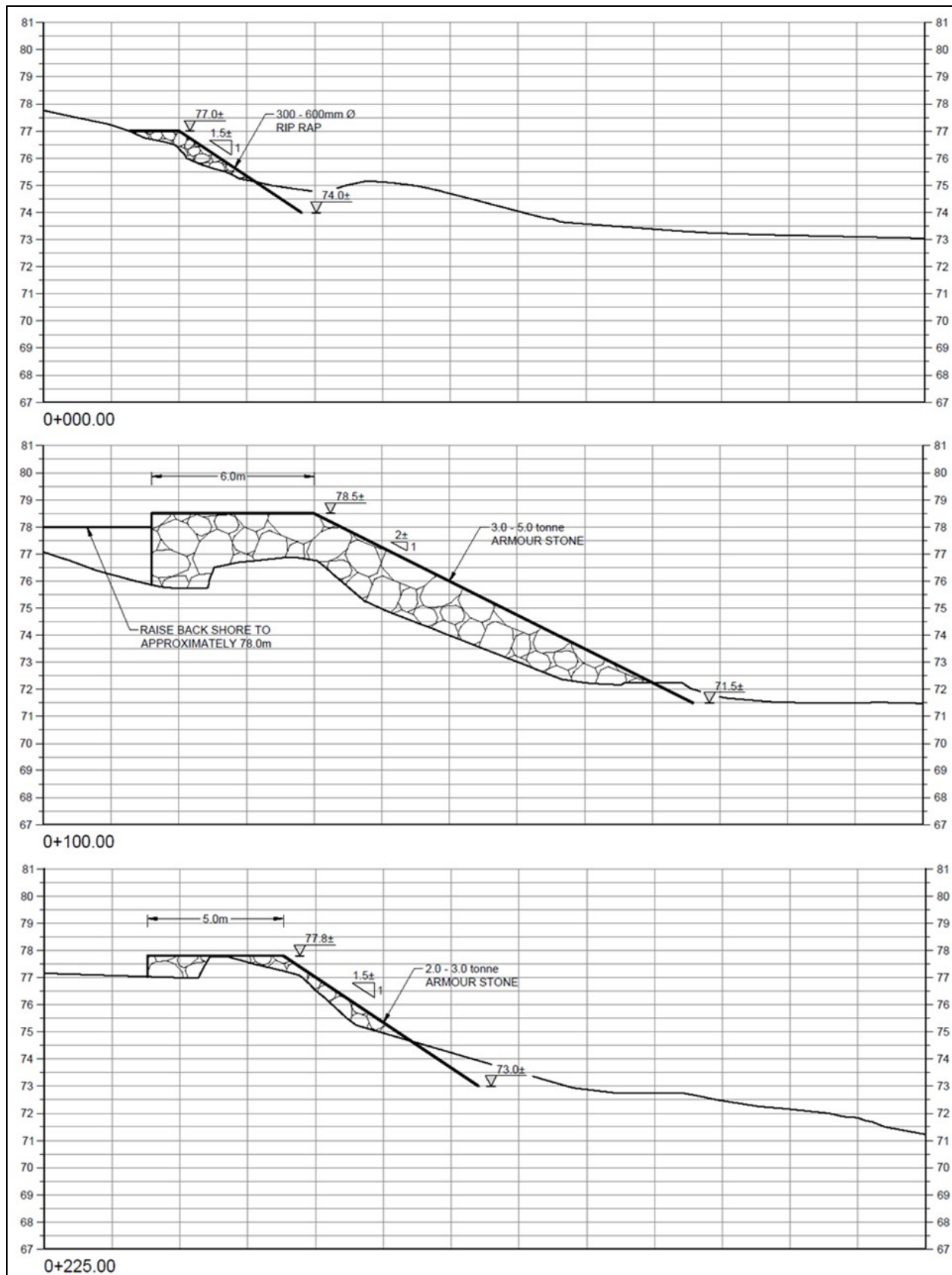


Figure 15. Sample cross-sections of WF 11.10. Source: Shoreplan, 2020.

At its eastern limit, the headland transitions into beach WF11.11. At this transition point, the concept design includes the creation of a berm, consisting of 2-3 tonne armourstone, to create a formal outlet area for the ponds (**Figure 16** and **Figure 17**). The elevation of the treatment will be 78.5 m.a.s.l. (**Figure 17**). The pond will drain into the lake through a plug of loosely placed armourstone that will have crevices to allow the water to filter through. The design was developed with consideration of maximum outflow conditions, which were calculated to be 1000 g/minute. This treatment will also facilitate the creation of a backwater or wetland area between the pond and the outlet since water will naturally pool behind the berm. The design also includes the presence of scattered stone in front of the berm to help hold in the toe of the adjacent beach (**Figure 18**).

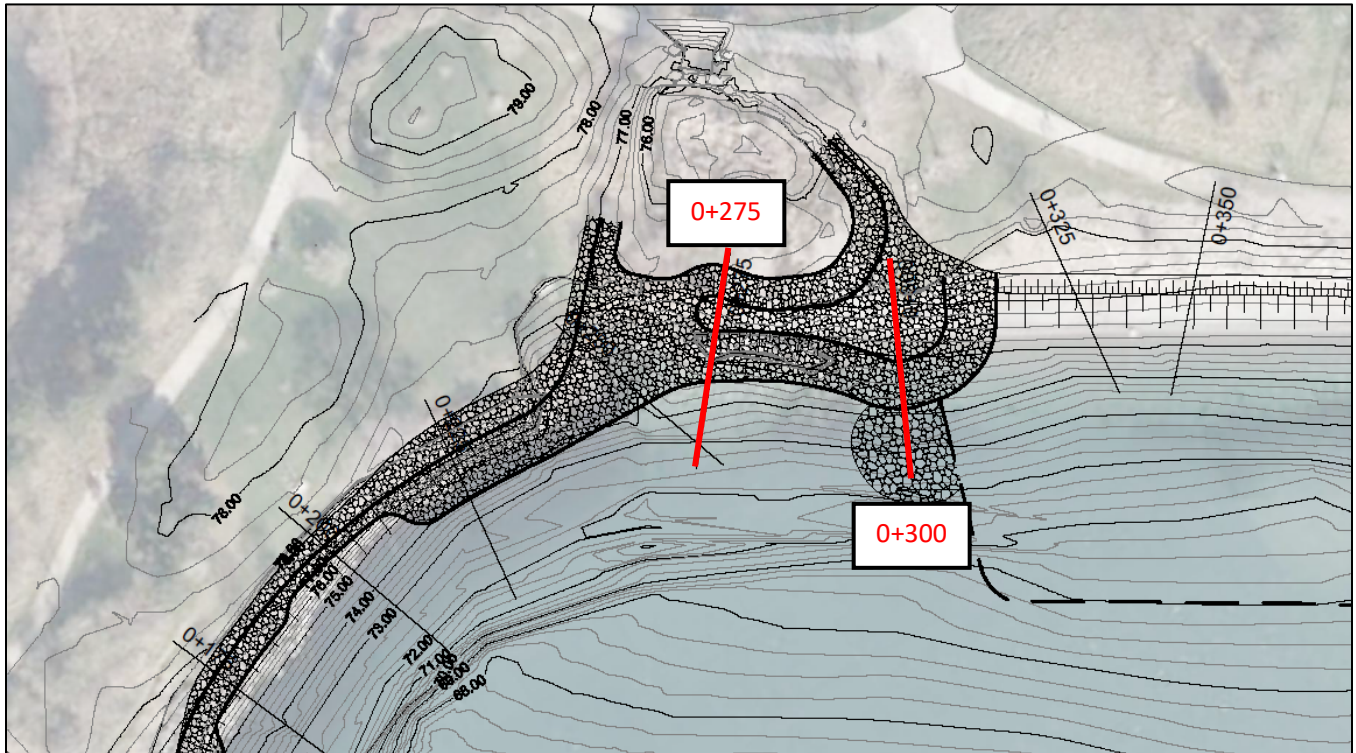


Figure 16. Concept design for the eastern extent of structure WF11.10. Source: Shoreplan, 2020.

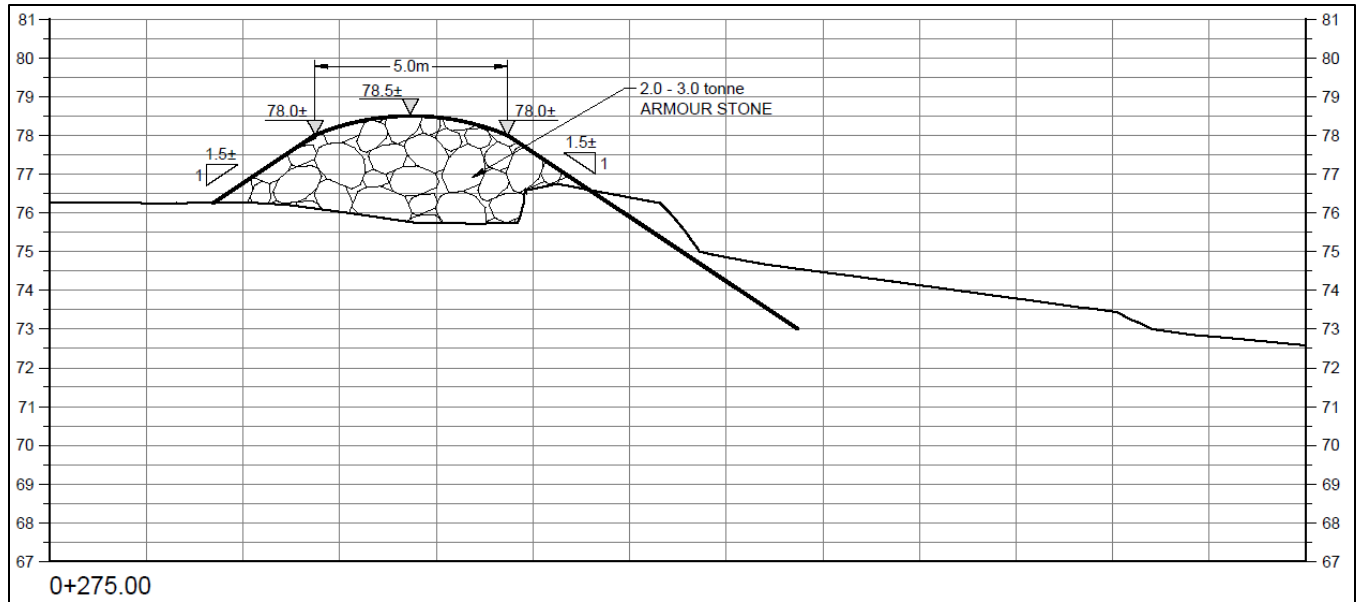


Figure 17. Cross-section of the berm at the eastern extent of WF11.10. Source: Shoreplan, 2020.

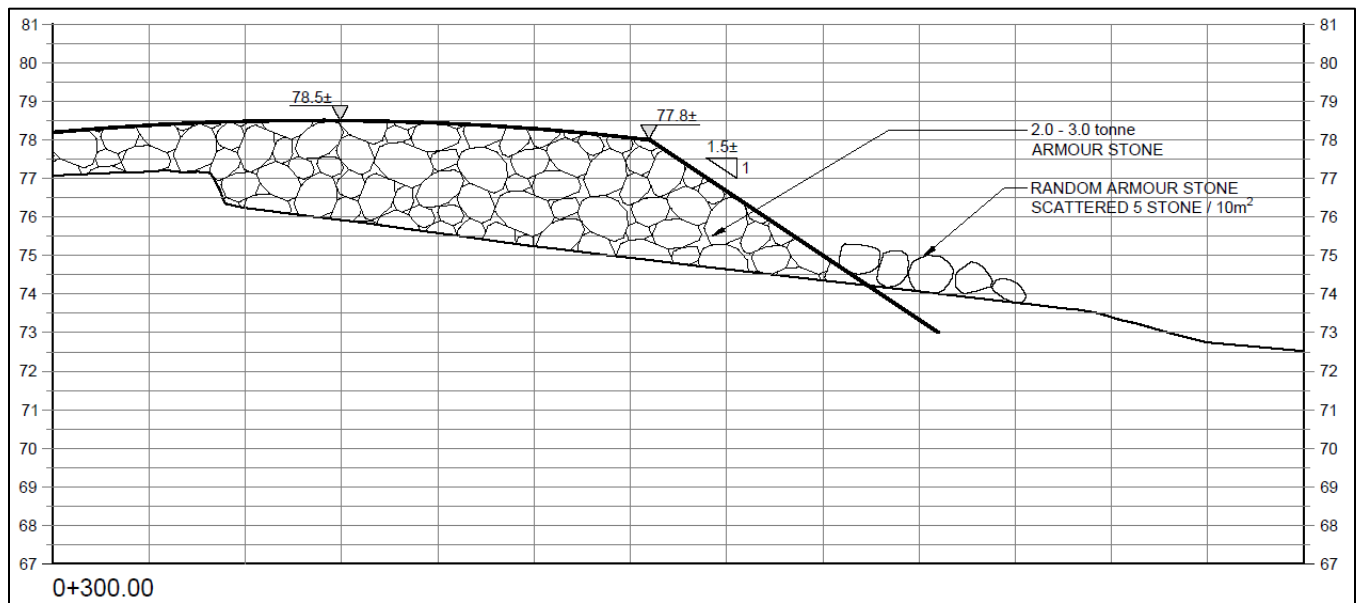


Figure 18. Cross-section of the berm at the eastern extent of WF11.10, with the scattered stone treatment. Source: Shoreplan, 2020.

5.2 WF11.11 – Cobble/Rubble Beach

The concept designs for the beach call for placement of fill over the existing profile of the beach (**Figure 19**). Four alternative options were proposed for the beach treatment which vary based on the cobble size and required quantity, ranging from 10 to 100 mm diameter stone (**Figure 20**). Profile details of the four options are summarized in **Table 3**.

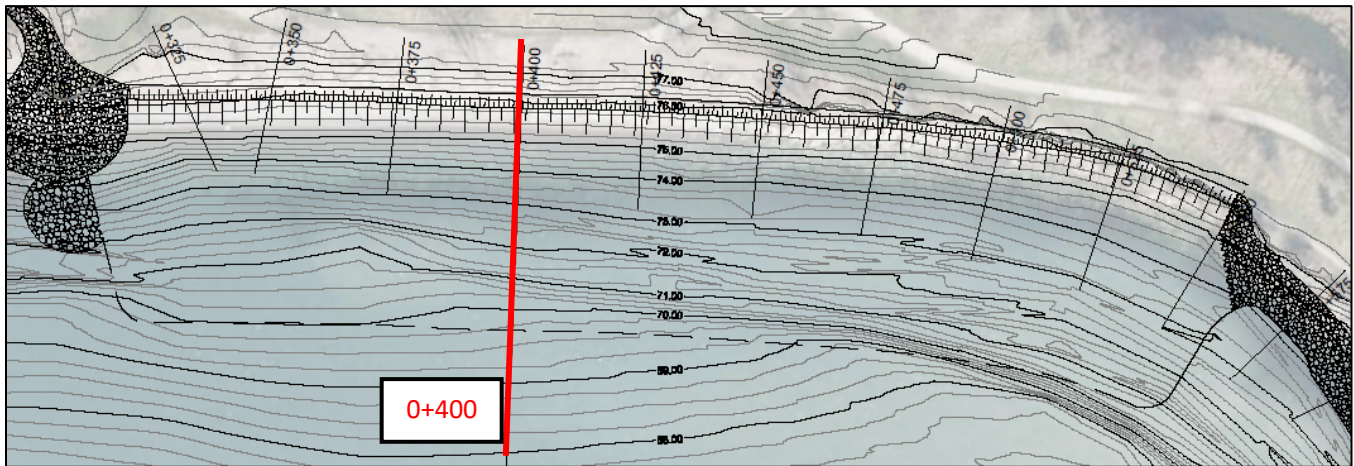


Figure 19. Concept designs for the beach (WF11.11). Source: Shoreplan, 2020.

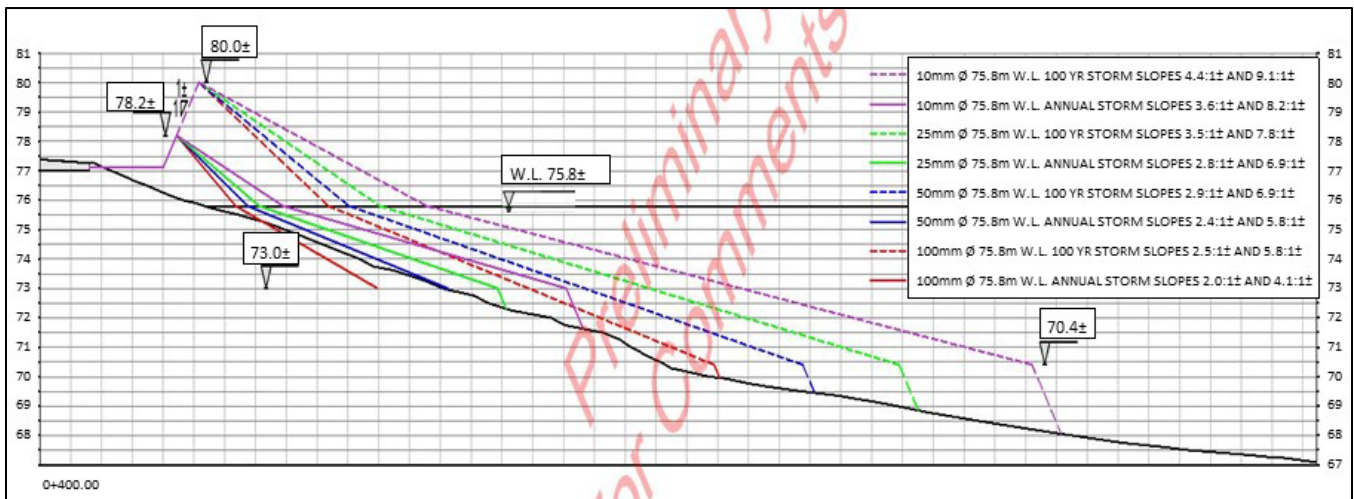


Figure 20. Cross-section of concept designs for WF11.11 with cobble size and profile options. Source: Shoreplan, 2020.

Table 3. Summary of beach profile designs based on selected cobble size. Source: Shoreplan, 2020.

Design	Cobble Size	Crest Elevation	Slope Above 75.8 m.a.s.l.	Slope Below 75.8 m.a.s.l.	Toe Elevation
A	10 mm	80 m.a.s.l.	4.4H:1V	9.1H:1V	70.4 m.a.s.l.
B	25 mm	80 m.a.s.l.	3.5H:1V	7.8H:1V	70.4 m.a.s.l.
C	50 mm	80 m.a.s.l.	2.9H:1V	6.9H:1V	70.4 m.a.s.l.
D	100 mm	80 m.a.s.l.	2.5H:1V	5.8H:1V	70.4 m.a.s.l.

In general, smaller cobble sizes will offer a more user-friendly beach environment for park users than larger stone, but more material will be required to sufficiently protect the shoreline. Treatments using smaller cobble will therefore have higher costs and a greater in-water footprint as well as greater potential for maintenance needs following large storm events. Conversely, larger cobble sizes will allow for a smaller in-water footprint, but will create a rougher surface and a less user-friendly beach environment. The overall cost for a larger cobble size will be lower with less frequent maintenance needs. The selection of the preferred cobble size will be made through discussions with TRCA, City of Toronto, and Shoreplan Engineering Limited and will consider overall cost, in-water footprint, maintenance requirements, user experience, and structure aesthetics.

The eastern limit of the beach is heavily damaged, as shown above in **Figure 8**, and its proposed treatment requires the extension of the eastern headland into the beach area (sections 0+550 to 0+600 on the drawings). This treatment is described in the following section with the treatment of the eastern headland (WF11.12).

5.3 WF11.12 – Eastern Headland

The concepts for the eastern headland involve placement of fill over most of the existing profile supplemented with an extension of the structure toward the beach for the heavily damaged area located between sections 0+550 to 0+600 on the drawings (highlighted in yellow on **Figure 21**). The proposed treatment for the reconstructed section (station 0+575) is shown in blue on the cross sections and includes the placement of rip rap as a core material capped with 2-3 tonne armourstone, as presented in **Figure 22**. The reconstruction will require excavation of the existing profile and placement of fill beneath or behind the structure. This treatment provides alternative options for crest elevation, crest width and slope positioning: the black option presents the highest and narrowest possible crest (78.5 m.a.s.l. and 3.0 m wide) while the green option presents the lowest and widest possible crest (77.5 m.a.s.l. and 7.0 m wide). The structure can be designed to these specifications or to any elevation in between. In general, if the crest is built to a lower elevation it will need to be built wider.

The proposed treatment for the rest of the headland (sections 0+625 to 1+225) includes placement of 3-5 tonne armourstone fill over the existing profile as shown in **Figure 22**, cross-sections 0+850 and 1+150. Similar to the treatment described above, alternative options for crest elevations and widths are provided for various sections, shown in green and black on the cross-sections. In general, a higher crest will require less excavation and a lesser impact on the backshore areas but may obscure lake sight lines. This would be mitigated in the design; fill would be used to raise the elevation of the backshore areas to ensure that there is a maximum differential no greater than 1m between the backshore and crest elevations. Wider crests would require more clearing and vegetation removal and will encroach further into parkland and park paths, however, sightlines would be maintained and it is less likely that raising backshore elevation would be necessary.

Further assessment of the backshore conditions, including locations of existing trails and presence of vegetation, will be conducted for each area. This will help determine the options that will have the least impact for each section. Additional discussions between TRCA, City of Toronto and Shoreplan Engineering Limited will be required in order to select the preferred options. In addition to the recommendations of the backshore assessment, cost, maintenance requirements and aesthetics will be key considerations throughout the selection process.

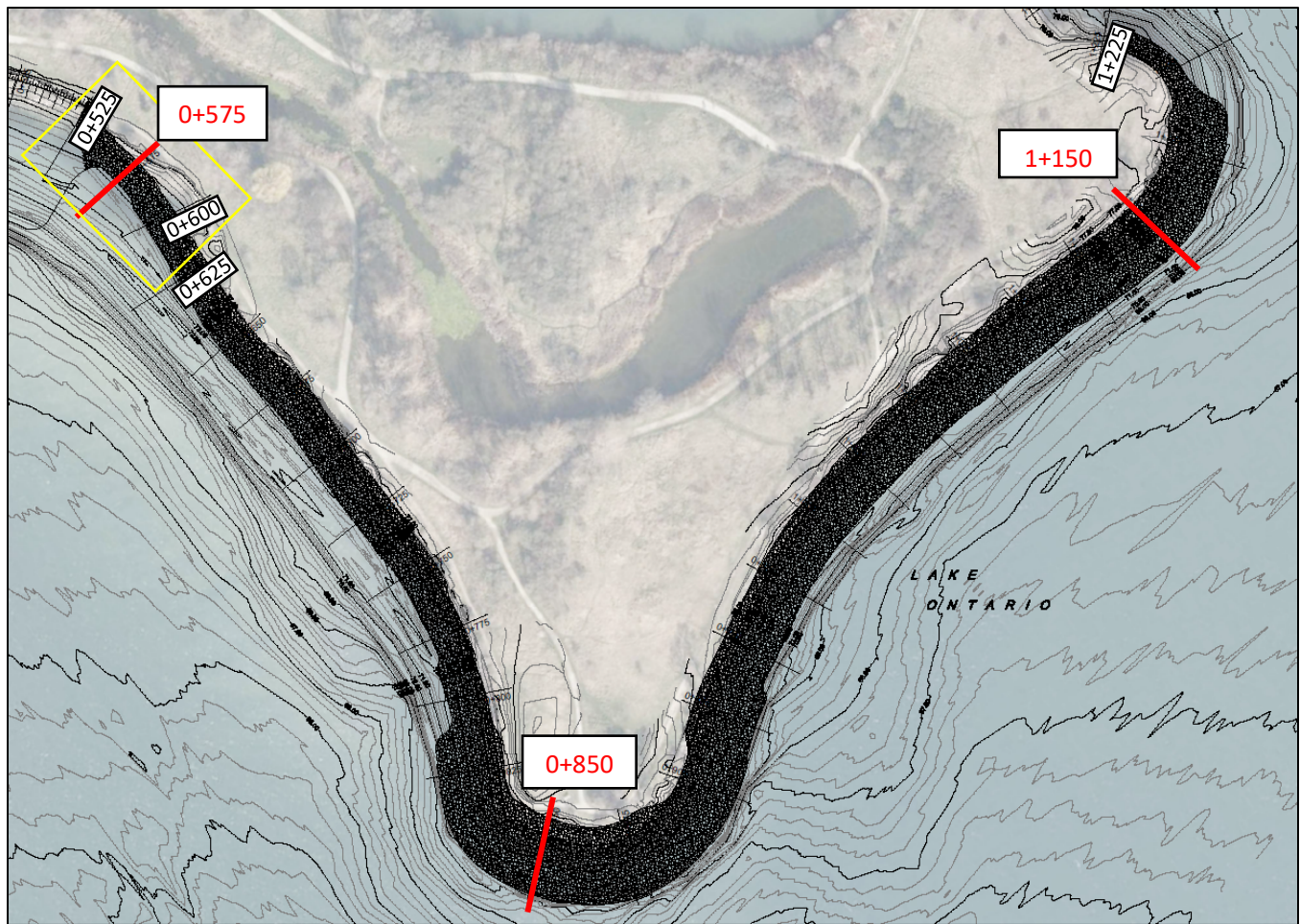


Figure 21. Planview of structure WF11.12. The yellow box indicates the approximate extents of the proposed reconstruction.
Source: Shoreplan, 2020.

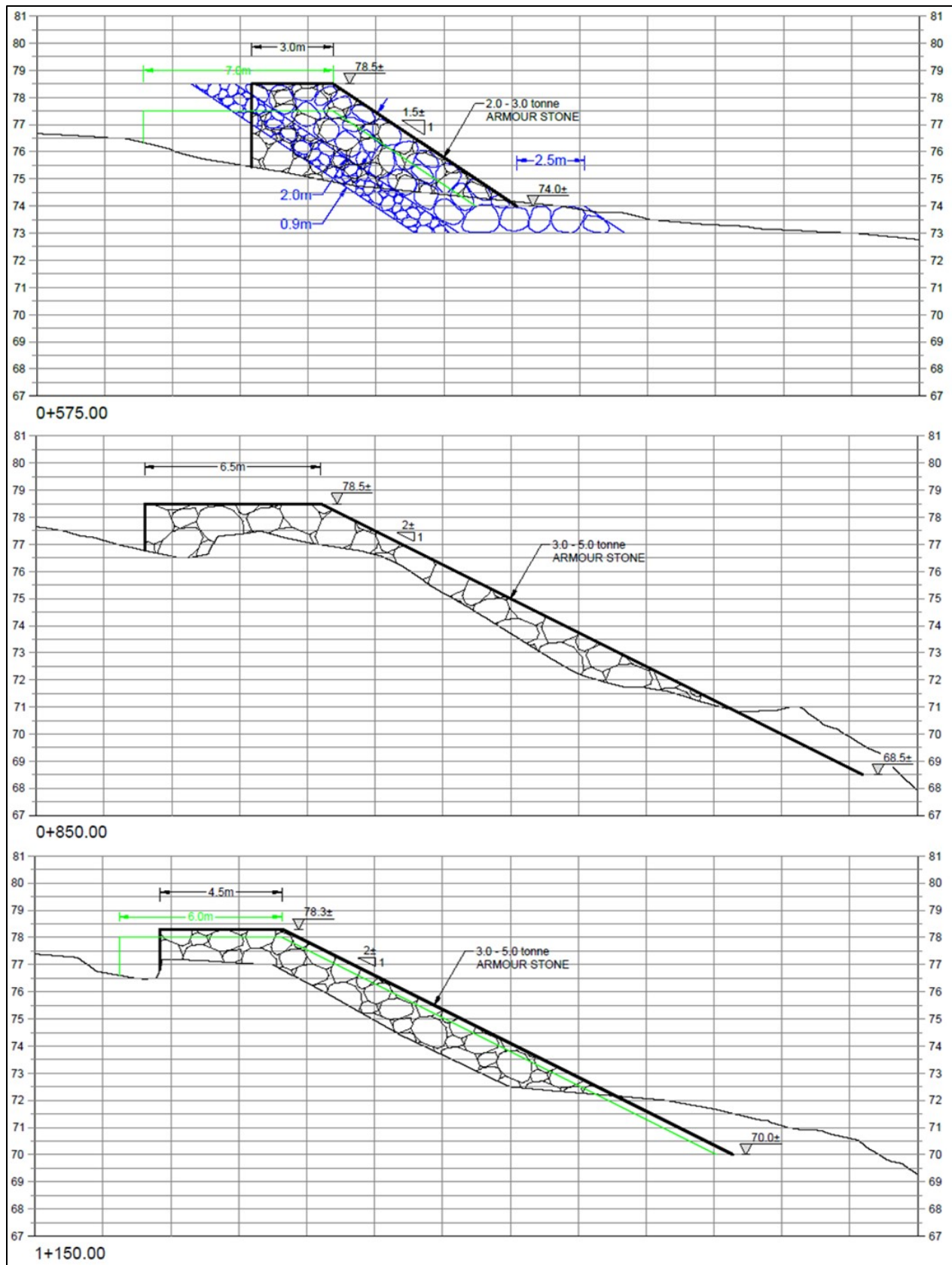
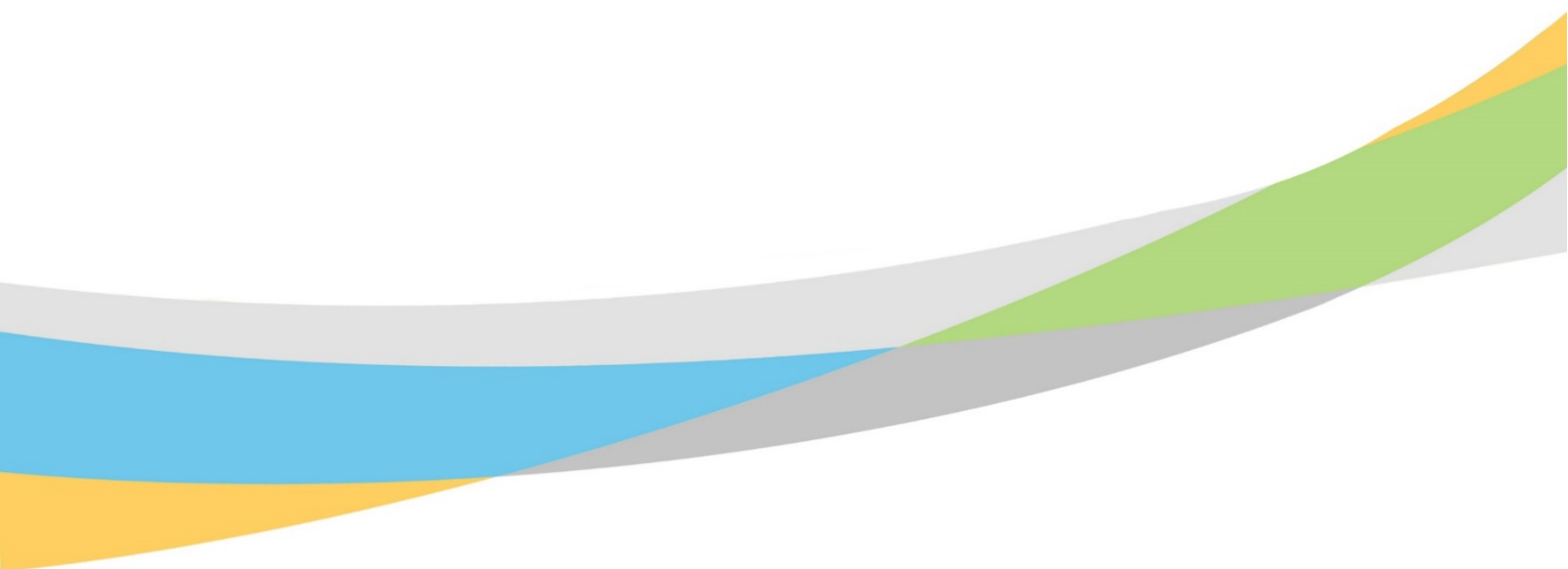


Figure 22. Sample cross-sections of structure WF11.12 with profile options in green. Source: Shoreplan, 2020.



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